

Computer Knowledge—Basic Level: Bits and Bytes

Subject: Computer Knowledge

Computers represent information using binary digits, or bits. A bit has one of two possible values, usually written as 0 or 1. These values can represent electrical states, logical conditions, or two possible signals. A group of eight bits is called a byte. Because each bit has two possibilities, one byte can represent 2^8 , or 256, different patterns. The patterns may be used for numbers, characters, colors, instructions, or parts of a file depending on the encoding system.

Binary place values are powers of two. From right to left, the places in an eight-bit number represent 1, 2, 4, 8, 16, 32, 64, and 128. The binary number 00000101 represents 5 because it contains 4 and 1. The binary number 00001010 represents 10 because it contains 8 and 2. Computers use electronic circuits called logic gates to perform operations on binary values. AND, OR, and NOT gates are basic examples. Larger circuits combine these operations to add numbers, compare values, store data, and execute instructions.

Bytes are also used to describe storage and file sizes. A kilobyte, megabyte, gigabyte, and terabyte represent increasingly large amounts of data. In everyday computing, these terms may be used approximately, while some technical contexts distinguish powers of 1024 from decimal multiples of 1000. A text character may use one byte in a simple encoding, but modern text encodings can use multiple bytes for characters from different writing systems and symbols. An image is stored as numerical information describing pixels, and sound is represented by sampled measurements of air pressure.

Memory and storage are not the same. Random-access memory, or RAM, temporarily holds data and instructions currently being used. Storage devices such as solid-state drives and hard disks retain data after the computer is turned off. A larger file usually requires more bytes, although compression can reduce size. Lossless compression preserves the original data exactly, while lossy compression removes some information to create a smaller file.

Bits and bytes also appear in networking. Internet speed is often measured in bits per second, while file size is usually measured in bytes. Since one byte contains eight bits, a speed quoted in megabits per second is not numerically the same as a speed in megabytes per second. Understanding this difference helps users estimate download times. The basic idea is simple: bits are the smallest binary units, bytes group eight bits, and larger units describe collections of bytes. Every application, from a calculator to a video service, ultimately stores and processes information as patterns of bits.